CSE5026: Foundation and Frontiers of Cognitive Science

Language and Syntax

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Overview

- Neural Basis of Language Processing
- Part-of-Speech Tagging（词性标注)
- Syntactic Parsing（句法解析)
- Cognitive Processing

Source: https://en.wikipedia.org/wiki/Broca%27s_area

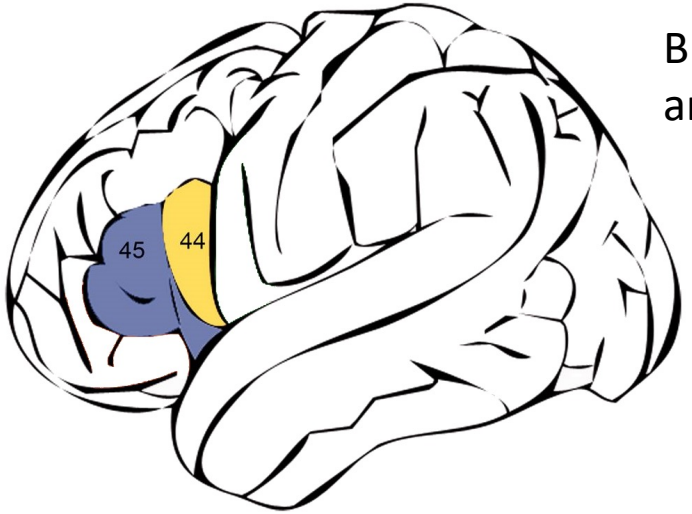
Neural Basis of Language

（失语症）

- French neurologist Broca's studies of *aphasia* linked speech production to the left inferior frontal gyrus, now called **Broca's area**. （左侧额下回）
- Broca area's function: speech planning and production.

（运动性/表达型失语症）

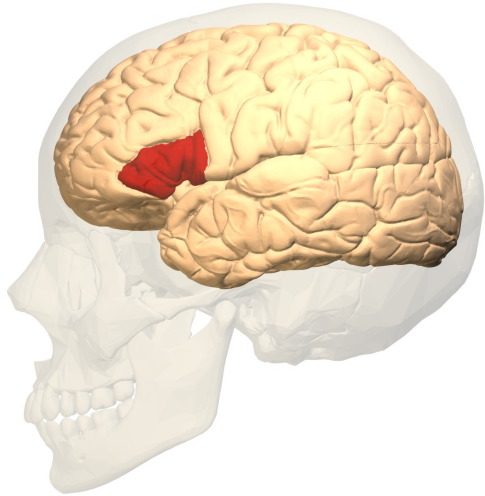
- Damage causes **expressive aphasia**:
  - Severely impaired speech, with preserved comprehension.
  - Slow, fragmented speech.



Brodmann areas 44, 45



Paul Pierre Broca (1824--1880)



Source: https://en.wikipedia.org/wiki/Broca%27s_area

Expressive Aphasia

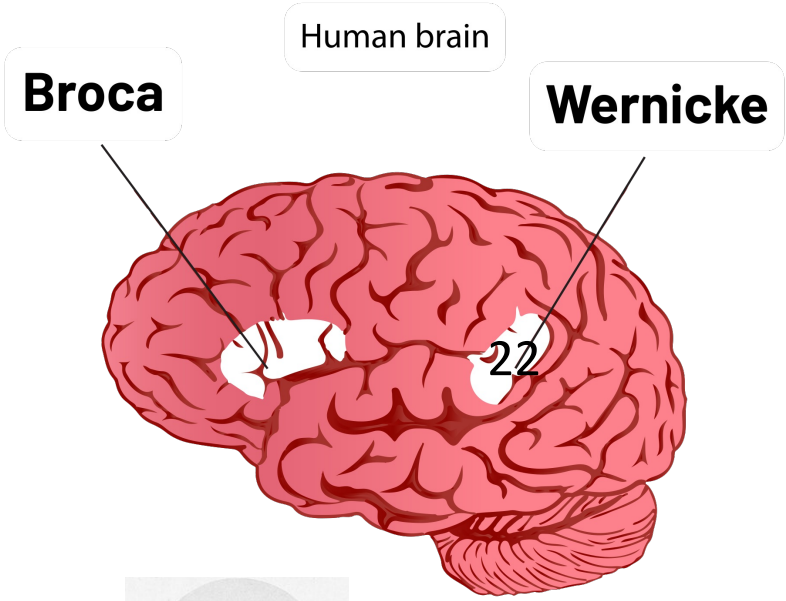
- Broca's patients:
  - LVL (Tan): He was able to repetitively produce only the word 'temps' (French word for "time").
  - Lelong: He could only say five words, 'yes', 'no', 'three', 'always', and 'lelo' (a mispronunciation of his own name).

> "Yes … ah … Monday … ah … father and Peter H. (his name) … and father … ah … hospital … and ah … Wednesday … Wednesday, nine o'clock … and, oh … Tuesday … ten o'clock … ah, doctor … two … doctors … and ah … teeth … yes."

- Modern example: Sarah Scott's case: Aphasia caused by stroke and recovery after five years.

Neural Basis of Language

- German neurologist and physician Wernicke identified **Wernicke's area**. Patients with damage here can speak fluently, but their sounds, words, and sentences lack meaning.
  - Located in Brodmann area 22 in the superior temporal gyrus. (颞上回)
  - Wernicke's area's function: understanding spoken and written language (comprehension).
  - Damage causes **fluent aphasia**: severely impaired comprehension, with natural speech rhythm and *relatively normal syntax*. (流利性失语症)





Carl Wernicke
(1848--1905)

Source: https://en.wikipedia.org/wiki/Wernicke%27s_area

Fluent Aphasia

- A patient with Wernicke‘s aphasia was asked what brought him to the hospital. His response was:

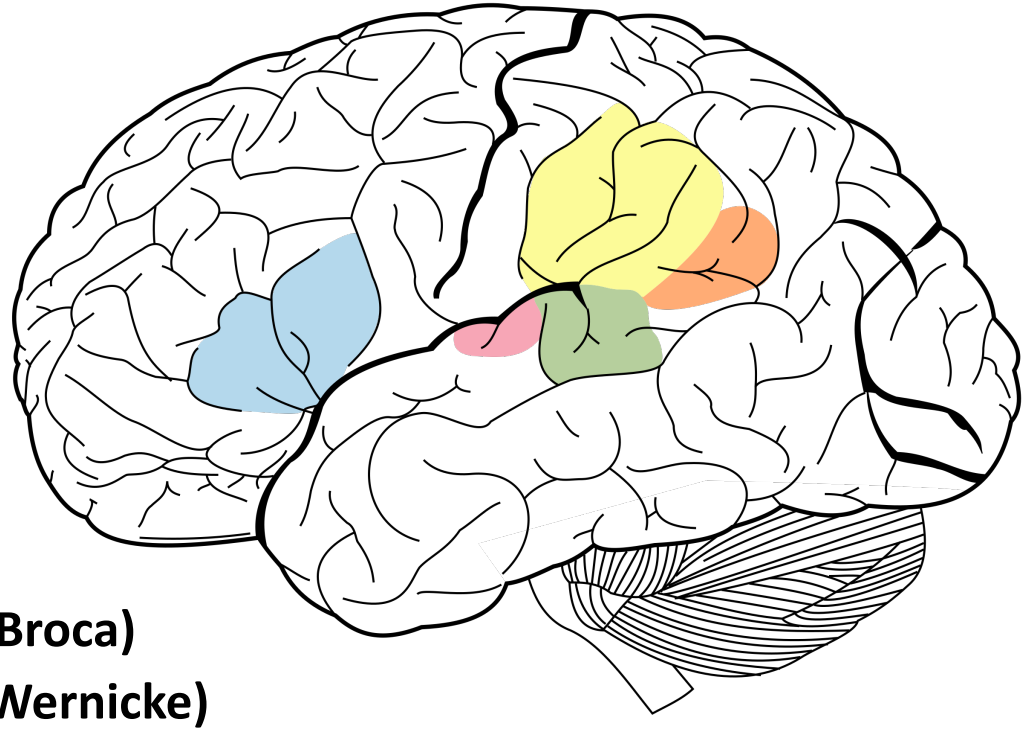
> Is this some of the work that we work as we did before? ... All right ... From when wine [why] I'm here. What's wrong with me because I ... was myself until the taenz took something about the time between me and my regular time in that time and they took the time in that time here and that's when the time took around here and saw me around in it's started with me no time and I bekan [began] work of nothing else that's the way the doctor find me that way...

- Patients often do not recognize their speech impairment because their ability to understand their own speech has changed.

Source: https://en.wikipedia.org/wiki/Language_center

Language Center

- Brain regions involved in speech and language processing.



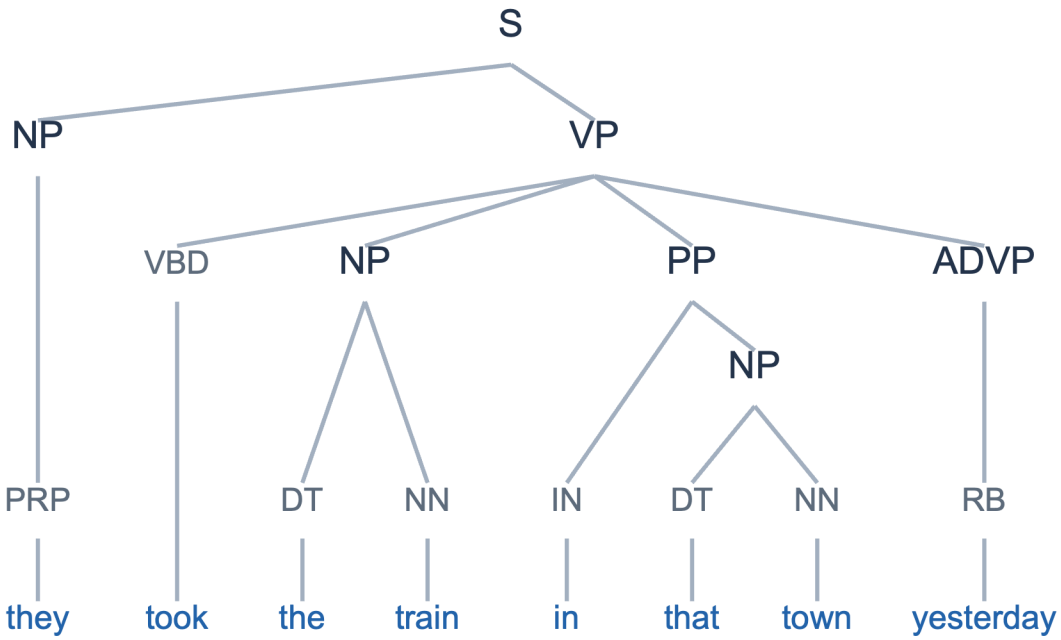
- **Blue**: Broca's area
- **Green**: Wernicke's area
- **Orange**: angular gyrus
- **Pink**: primary auditory cortex
- Yellow: supramarginal gyrus
  （顶外侧回）

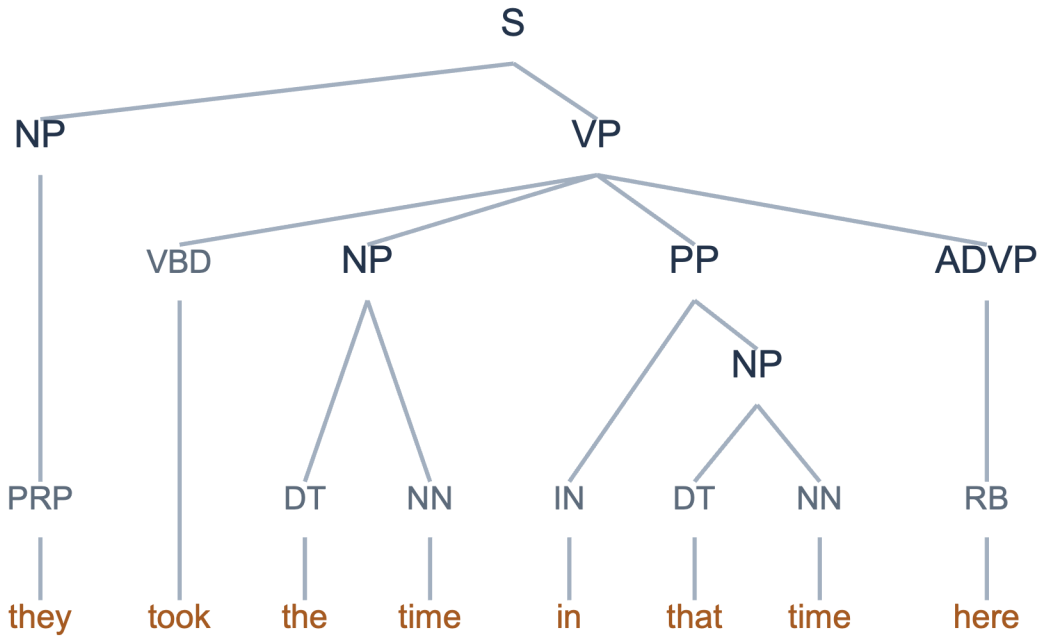
Main functions:

- **Speech production (Broca)**
- **Auditory language (Wernicke)**
- **Visual language (Wernicke, angular gyrus)**
  （角回）

How is it related to syntax?

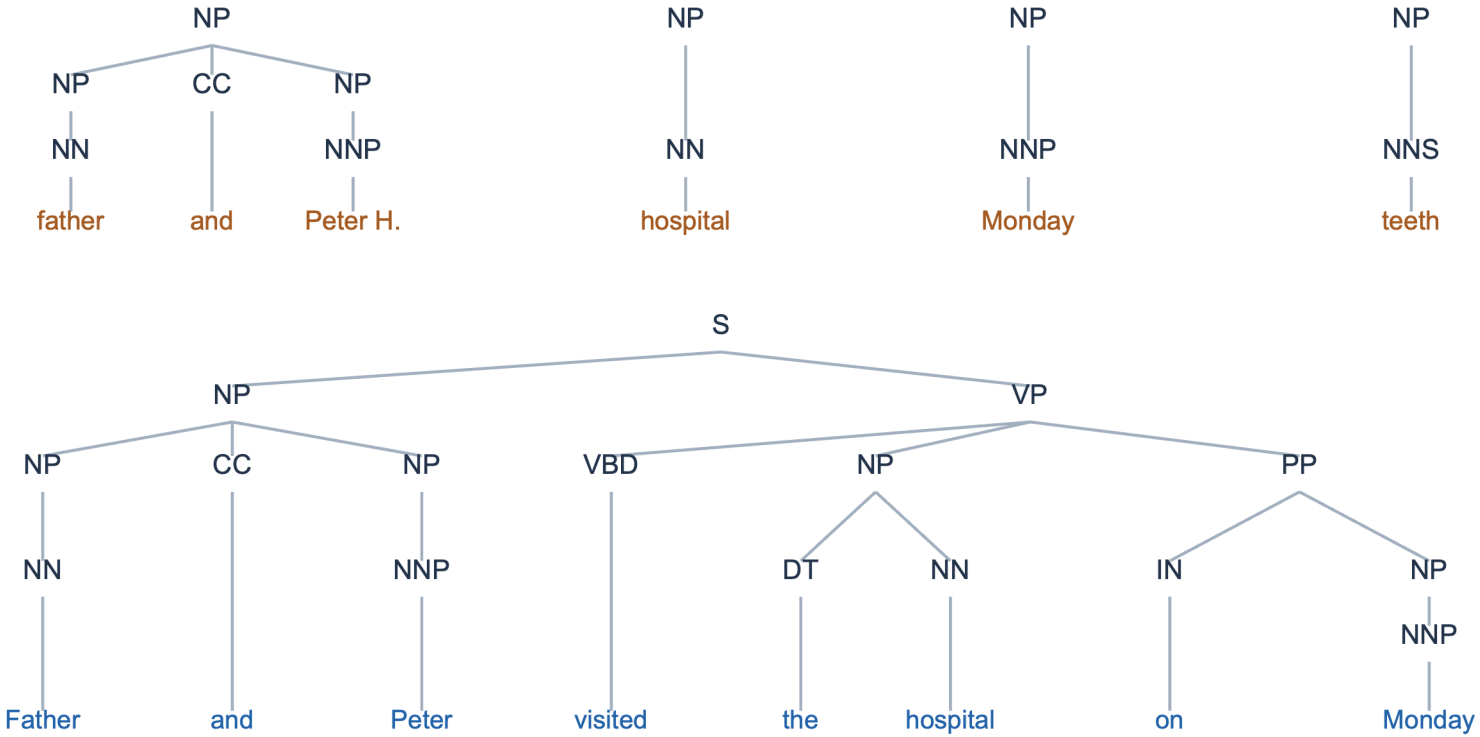
- A Fluent aphasia patient: *they took the time in that time here
- Normal person: they took the train in that town yesterday





How is it related to syntax?

- An expressive aphasia patient: *father and Peter H. ... hospital, Monday, teeth
- Normal person: father and Peter visited the hospital on Monday

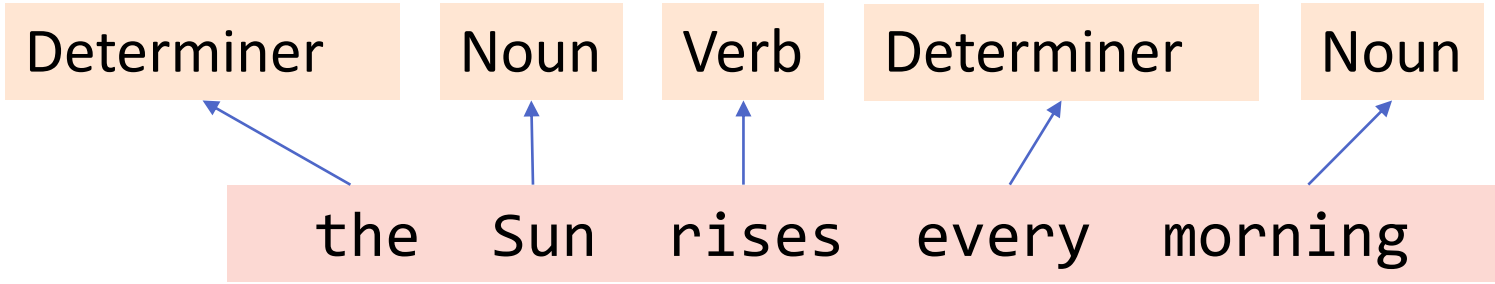


Overview

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- **Part-of-Speech Tagging（词性标注）**
- Syntactic Parsing（句法解析）
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Part of Speech

- Words can be classified into grammatical categories



determiner: e.g., a, the

"n" for Noun ☺

Results returned from: https://corenlp.run/

- These categories are known as **part of speech (POS, POS tags)**, word classes, or simply grammatical categories.

POS in Chinese Language

Not a complete list

实词 Notional Words or **content** words

名词 Nouns	动词 Verbs	助动词 Auxilliary Verbs	形容词 Adjectives	数词 Numerals	量词 Measure Words		代词 Pronouns		
					名量词 Nominal	动量词 Verbal	人称代词 Personal	指示代词 Demonstrative	疑问代词 Interrogative
国家、妹妹、玫瑰、颜色、月亮	工作、学习、走、吃、打	应该、可以、能、会、想	漂亮、诚实、慢、坏、红	一、二、百、万、亿	个、条、张、公分、只	次、遍、回、趟、轮	我们、你、他、她、咱们	这、那、各、每、该	什么、啥、哪、谁、怎么

unique in Chinese

虚词表 Functional Words

副词 Adverbs	介词 Prepositions	连词 Conjunctions	助词 Particles			叹词 Interjections	象声词 Onomatope
			结构助词 Structural	动态助词 Aspectual	语气助词 Modal		
很、都、就、也、已经	从、向、在、被、把	跟、但是、或者、并且、因为	的、地、得	了、者、过	吗、吧、呢、了	喂、哎呀、嗯、哦	哗哗、乒乓

source: https://chinesenotes.com/grammar_intro.html

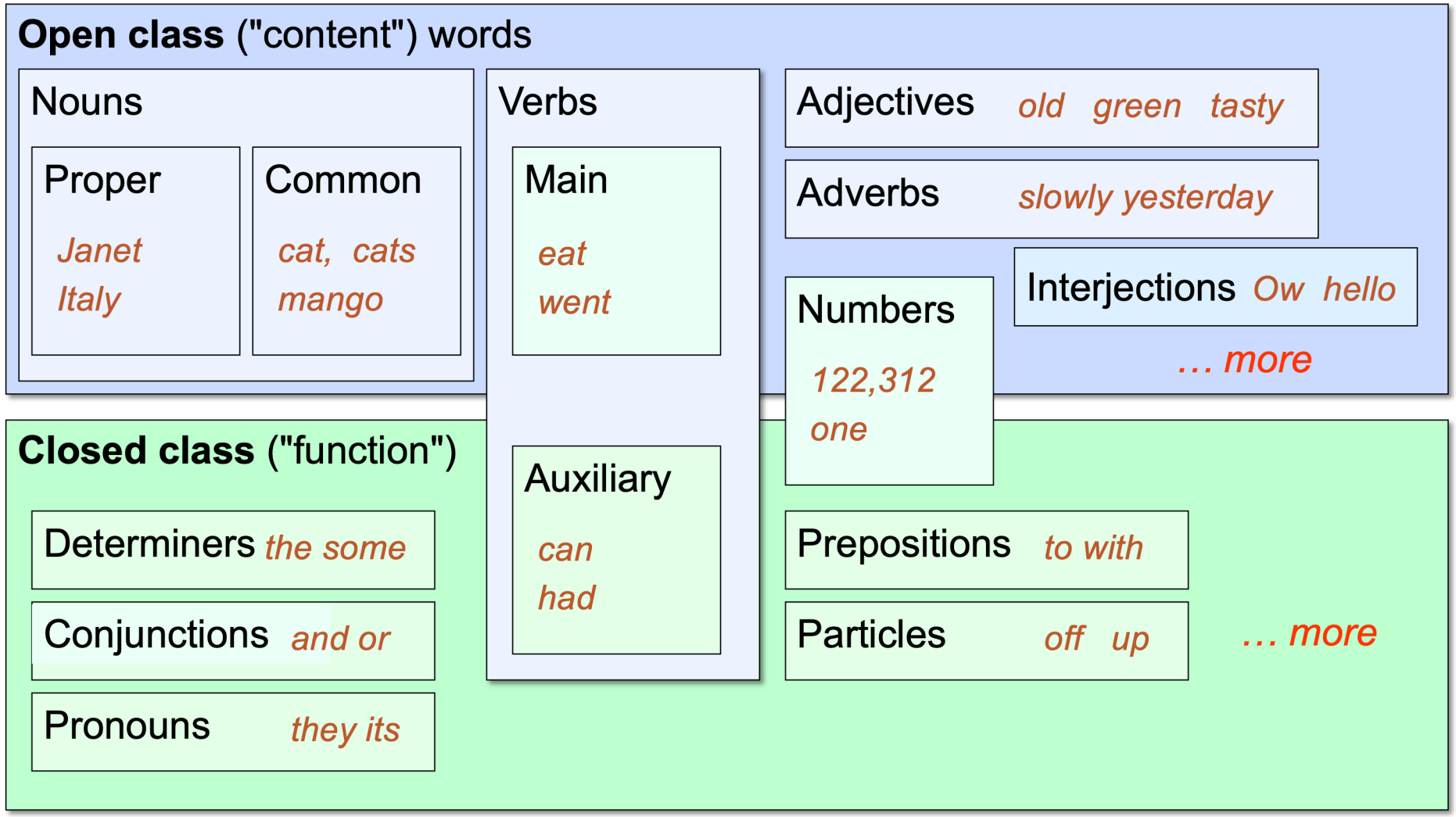
Two classes of words: Open vs. Closed

- Open class
  - Usually **content** words: **Nouns, Verbs, Adjectives, Adverbs**
  - New nouns and verbs like *AI* or *GPT* … continuously being created/borrowed
- Closed class
  - Relatively fixed membership
  - Usually **function** words: short, frequent words with grammatical function
    - **determiners: a, an, the**
    - **pronouns: she, he, I**
    - **prepositions: on, under, over, near, by, …**

Adapted from: https://web.stanford.edu/~jurafsky/slp3/slides/8_POSNER_intro_May_6_2021.pptx

Common POS tags in English

Adapted from: https://web.stanford.edu/~jurafsky/slp3/slides/8_POSNER_intro_May_6_2021.pptx



How difficult is POS tagging in English?

（词表）

- Roughly 15% of word ***types*** (in the vocabulary) are ambiguous
- Hence 85% of word types are unambiguous
- E.g., *James* is always PROPN, *healthy* is always ADJ
- But those 15% tend to be very common (frequet). So ~60% of word ***tokens*** (in the data) are ambiguous

（训练数据）

Some ambiguous examples:

Love loves to love love.
NP VP VP NP

-- *Ulysses*
JAMES JOYCE

Source: https://web.stanford.edu/~jurafsky/slp3/slides/8_POSNER_intro_May_6_2021.pptx

Difficulty of Automatic POS Tagging (in English)

- How many tags are correct? (Tag accuracy)
- About **97%**
  - Hasn‘t changed in the last 10+ years; **HMMs**, **CRFs**, BERT perform similarly
  - Human accuracy about the same
- But baseline is **92%**!
  - Baseline: stupidest possible method -- "Most frequent class baseline"
    - Tag every word with its most frequent tag (and unknown words as *nouns*)

Source: https://web.stanford.edu/~jurafsky/slp3/slides/8_POSNER_intro_May_6_2021.pptx

POS Tagging in Chinese

- Similar performance as English

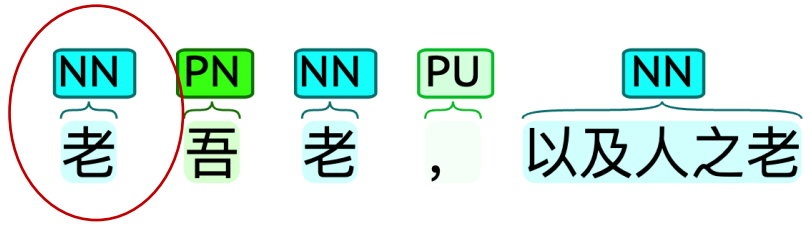
System	F1 score
Tian el. al. (2020)	96.92
Meng et. al. (2019) (Glyce + BERT)	96.61
Meng et. al. (2019) (BERT)	96.06
Shao et. al. 2017	94.38

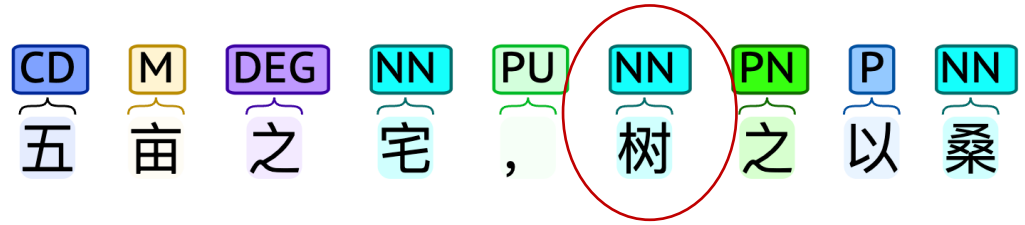
Data source: https://chinesenlp.xyz/docs/pos_tagging.html

Example: “The quick brown fox jumped over the lazy dog.”

- 快速 的 棕色 狐狸 跳过 了 懒惰 的 狗
- [快速] VA [的] DEC [棕色] NN [狐狸] NN [跳过] VV [了] AS [懒惰] VA [的] DEC [狗] NN

Maybe true for modern Chinese (with better labeled data), but not necessarily true for all cases:





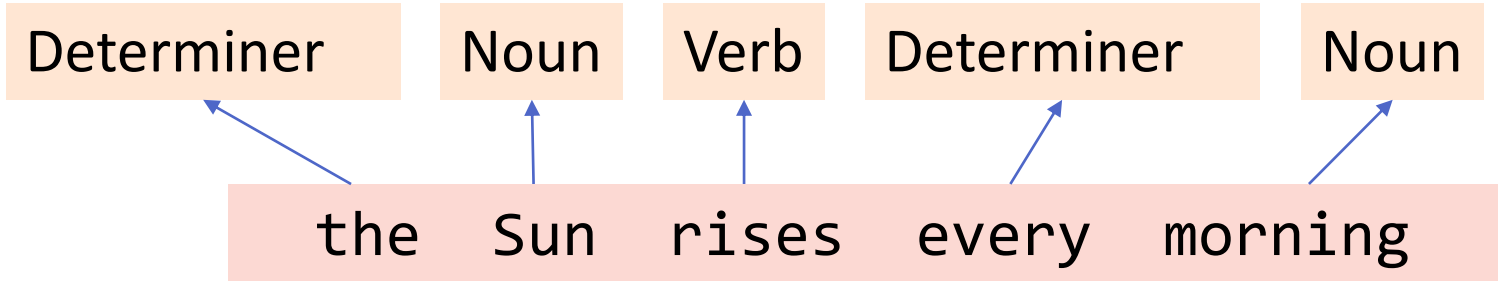
Noun or Verb?

Overview

- Neural Basis of Language Processing
- Part-of-Speech Tagging
- **Syntactic Parsing**
- Cognitive Processing

Grammar ≈ Syntax

- Greek: *syntaxis*, meaning “setting out together”
- The way words are arranged together
- Syntax: the rules governing sentence structure
- Part-of-speech is already a kind of simple syntactic notions



Constituency （成分）

- Groups of words can behave as **single units**, or **constituents**
- Ex. noun phrases (***NP***, sequence of words surrounding at least one noun)

Harry the Horse	a high-class spot such as Mindy's
the Broadway coppers	the reason he comes into the Hot Box
they	three parties from Brooklyn

- Why they form constituents? (the following NPs all appear before verbs)

three parties from Brooklyn *arrive*...
a high-class spot such as Mindy's *attracts*...
the Broadway coppers *love*...
they *sit*

Invalid sentences:

*from *arrive*...	*as *attracts*...
*the *is*...	*spot *sat*...

Terminology

（词组）
- **Phrase**: a group of words that functions as a grammatical unit

（名词词组）
- **Noun phrase**: a phrase headed by a noun. Prepositional phrase: a phrase headed by a preposition

（专有名词）
- **Proper noun**: the name of a specific person, place, or entity

- **Nominal**: a noun-like word or expression

（名词性成分）

How can constituency describe grammar?

- How do we describe rules such as "*Noun Phrases can occur before verbs*", "*Prepositional Phrases can occur after verbs*" and so on?
- Need a more formal way!
- **C**ontext-**F**ree **G**rammar (**CFG**) 上下文无关语法

（上下文无关语法）

Context-Free Grammars (CFG)

Wundt, German psychologist
Chomsky, American linguist
Backus & Naur, American computer scientists (FORTRAN, ALGO)

- Also called Phrase-Structure Grammar
- Wilhelm Wundt (1900), Chomsky (1956), Backus (1959), Naur (1963)
- A set of **rules** or **productions** that express how symbols can be grouped and ordered.

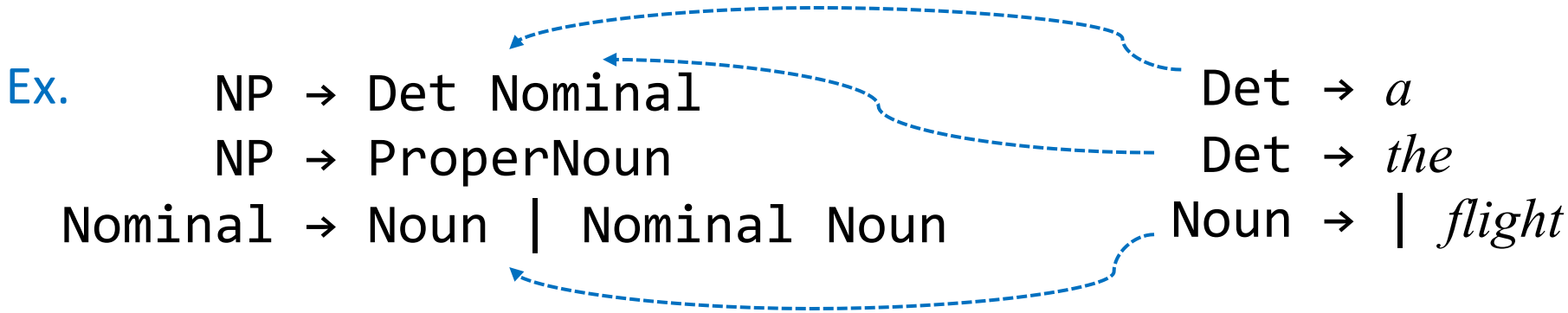
Ex.

```
NP → Det Nominal
NP → ProperNoun
Nominal → Noun | Nominal Noun
```

Meaning: An NP (noun phrase) can consist of
- either a proper noun
- or a determiner (Det) followed by a nominal

A nominal can contain one or more nouns

Rules can be hierarchically embedded



Two classes of symbols: **terminal** and **non-terminal**

- **terminal:** symbols that are actual words in language (“a”, “the”, …)
- **non-terminal**: symbols that express abstractions over terminals

CFG as a device for generating sentences

Rules:

① NP → Det Nominal
② NP → ProperNoun
③ Nominal → Noun | Nominal Noun

④ Det → *a*
⑤ Det → *the*
⑥ Noun → | *flight*

Start from *NP*

Use rule ① to rewrite *NP* as: *Det Nominal*

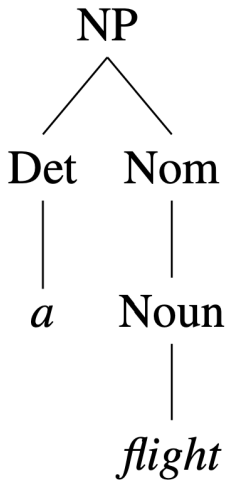
Use rule ③ to rewrite *Nominal* as: *Noun*

Use rule ④ to rewrite *Det* as: *a*

Use rule ⑥ to rewrite *Noun* as: *flight*

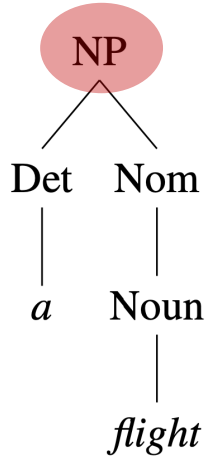
（解析树）

Result: **Parse tree**



CFG: More terms and rules

NP **dominates** all nodes in the tree



A grammar must have one designated **start symbol**, often called ***S***.

Common rules:

A sentence can consist of a noun phrase followed by a verb phrase

`S → NP VP` *I prefer a morning flight*

A VP can consist of a verb followed by assorted things

`VP → Verb NP` *prefer a morning flight*

Or VP may have a verb followed by a prepositional phrase

`VP → Verb PP` *leaving on Thursday*

A PP generally has a preposition followed by a noun phrase

`PP → Preposition NP` *from Los Angeles*

CFG: Examples from AITS corpus

A sample of grammar rules $\mathcal{L}_0$

Grammar Rules			Examples
S	$\rightarrow$	*NP VP*	I + want a morning flight
NP	$\rightarrow$	*Pronoun*	I
	|	*Proper-Noun*	Los Angeles
	|	*Det Nominal*	a + flight
Nominal	$\rightarrow$	*Nominal Noun*	morning + flight
	|	*Noun*	flights
VP	$\rightarrow$	*Verb*	do
	|	*Verb NP*	want + a flight
	|	*Verb NP PP*	leave + Boston + in the morning
	|	*Verb PP*	leaving + on Thursday
PP	$\rightarrow$	*Preposition NP*	from + Los Angeles

Air Travel Information System (ATIS) pilot corpus https://aclanthology.org/H90-1021/

Use ATIS grammar to generate "AITS-language"

S → NP VP Start with S, expand it with NP VP

Choose a random expansion of NP: *I*

NP →	*Pronoun*	I
|	*Proper-Noun*	Los Angeles
|	*Det Nominal*	a + flight

Choose a random expansion of VP: *Verb NP*

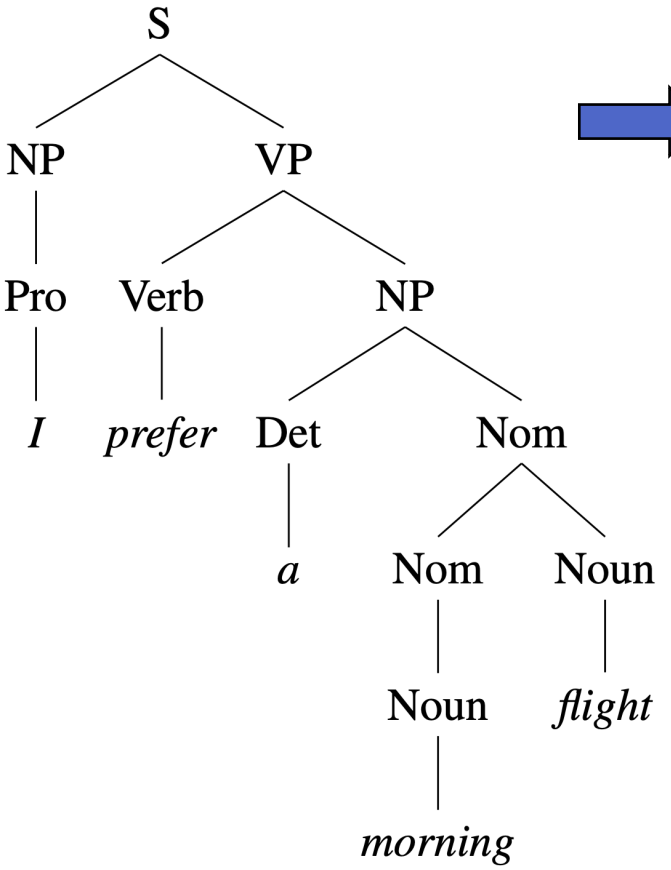
VP →	*Verb*	do
|	*Verb NP*	want + a flight
|	*Verb NP PP*	leave + Boston + in the morning
|	*Verb PP*	leaving + on Thursday

Verb NP

- want + a flight
- want + a morning flight
- *prefer + a morning flight*
- ……

I prefer a morning flight

Syntactic Tree: Bracketed Notation




```
(S (NP (PRO I)
   (VP (V prefer)
       (NP (Det a) (Nom (N morning) (N flight))))))
```

- Sentences that can be derived by a grammar are called **grammatical;** （语法正确）
- those cannot be derived are referred to as **ungrammatical**. （语法错误）

This distinguishment is vague, and only a very *simplified* model of how natural language really work.

Chomsky normal form (CNF)



Noam Chomsky, "the father of modern linguistics"

（标准形式）

- It is sometimes useful to have a **normal form** for grammars
- A grammar is in Chomsky normal form (CNF) if each production is

either $A \rightarrow B\ C$ or $A \rightarrow a$

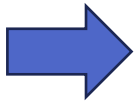
- That is, the right-hand side has ***two*** **non-terminals** or ***one*** **terminal**
- Any context-free grammar can be converted to a (weakly) equivalent CNF

Ex. $A \rightarrow B\ C\ D$ ⇒ $A \rightarrow B\ X$, $X \rightarrow C\ D$

Why CNF?

- Result in **binary branching** parse trees （二叉树）
- Produce smaller grammars

```
VP → VBD NP PP
VP → VBD NP PP PP
VP → VBD NP PP PP PP
VP → VBD NP PP PP PP PP
…
```

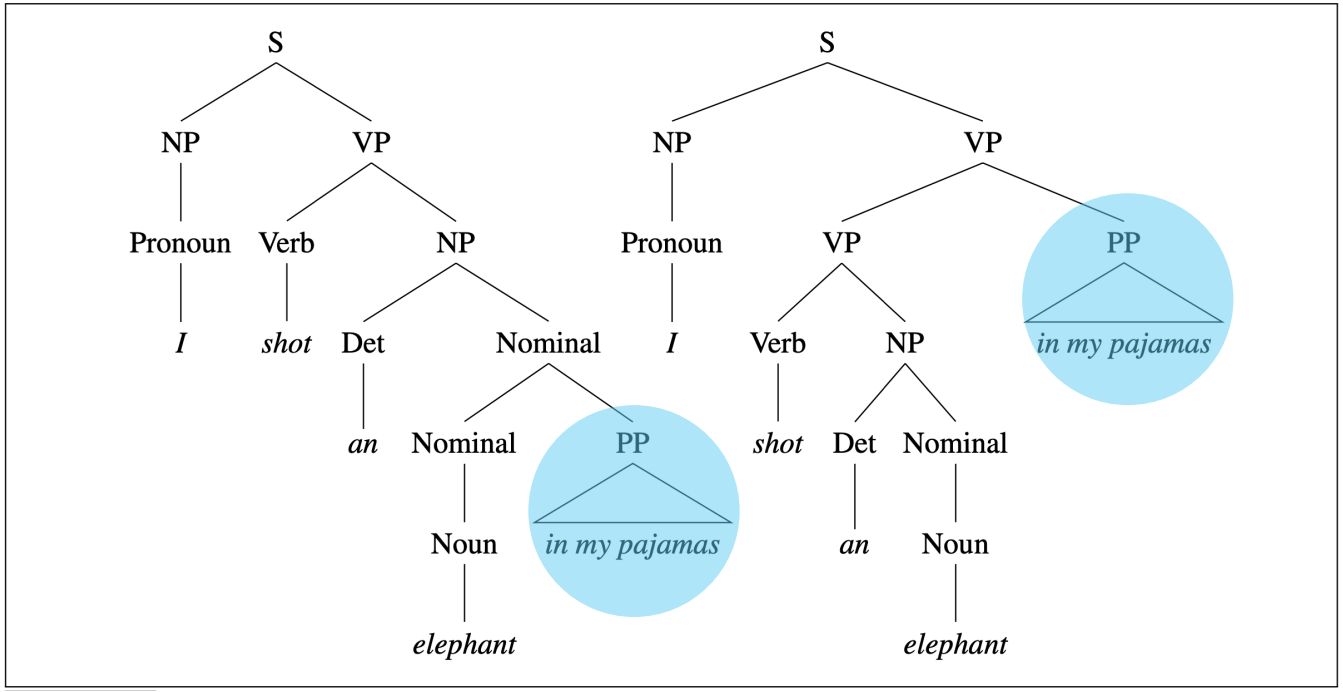


```
VP → VBD NP PP
VP → VP PP
```

Problem in syntactic parsing: Ambiguity (歧义)

- Structural ambiguity occurs when the grammar can assign more than one parse to a sentence. (一句话对应多个可能的解析树)

I shot an elephant in my pajamas



Left: elephant is in pajamas

Right: the action of shooting is done by me in my pajamas

PP-attachment ambiguity: the preposition phrase can be attached either as part of the NP or as part of the VP.

(PP: 介词短语)

Need a way to deal with these syntactic disambiguation

Parsing: A Dynamic Programming Approach

- **Dynamic programming (DP)** provides a powerful algorithmic framework for addressing the **ambiguity** problem
- A **DP** approach fills in a table of solutions to subproblems
- The complete table has the solution to all the subproblems
- In the case of parsing: the subproblems represent parse trees for all the constituents detected in the input

Parsing: A Dynamic Programming Approach

- **Basic assumption**: the **context-free nature** of grammar rules -- once a constituent has been discovered in a segment of input, we can store it and make available for use in any subsequent derivation that might require it
- CKY algorithm (Cocke-Kasami-Younger, 1961)

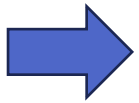
CKY Algo. Preprocessing: Conversion to CNF

- CKY algorithm requires grammars to first be in Chomsky Normal Form (CNF)
- I.e., the right-hand side of each rule must expand either to **two non-terminals** or to a **single terminal**

either $A \rightarrow B\ C$ or $A \rightarrow a$

- The remedy for rules that mix terminals and non-terminals is to introduce a new **dummy non-terminal**

INF-VP → to VP



INF-VP → TO VP
TO → to

CKY Algo. Preprocessing: Conversion to CNF

- For rules with right hand side longer than 2, introduce new non-terminals:

$A \rightarrow B\ C\ \gamma$ ⇒ $A \rightarrow X1\ \gamma$, $X1 \rightarrow B\ C$

Ex.

S → Aux NP VP ⇒ *S → X1 VP*, *X1 → Aux NP*

- For rules with a single non-terminal on the right (unit productions), rewrite the right hand side with non-unit production rules that they ultimately lead to

Nominal → Noun
Noun → book | flight

⇒

Nominal → book | flight

Grammar in CNF: Example

Grammar $\mathcal{L}_1$ (with adds to $\mathcal{L}_0$)

Grammar	Lexicon
$S \rightarrow NP\ VP$	$Det \rightarrow that \mid this \mid the \mid a$
$S \rightarrow Aux\ NP\ VP$	$Noun \rightarrow book \mid flight \mid meal \mid money$
$S \rightarrow VP$	$Verb \rightarrow book \mid include \mid prefer$
$NP \rightarrow Pronoun$	$Pronoun \rightarrow I \mid she \mid me$
$NP \rightarrow Proper\text{-}Noun$	$Proper\text{-}Noun \rightarrow Houston \mid NWA$
$NP \rightarrow Det\ Nominal$	$Aux \rightarrow does$
$Nominal \rightarrow Noun$	$Preposition \rightarrow from \mid to \mid on \mid near \mid through$
$Nominal \rightarrow Nominal\ Noun$	
$Nominal \rightarrow Nominal\ PP$	
$VP \rightarrow Verb$	
$VP \rightarrow Verb\ NP$	
$VP \rightarrow Verb\ NP\ PP$	
$VP \rightarrow Verb\ PP$	
$VP \rightarrow VP\ PP$	
$PP \rightarrow Preposition\ NP$	

Grammar $\mathcal{L}_1$ in CNF

$\mathcal{L}_1$ in CNF
$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

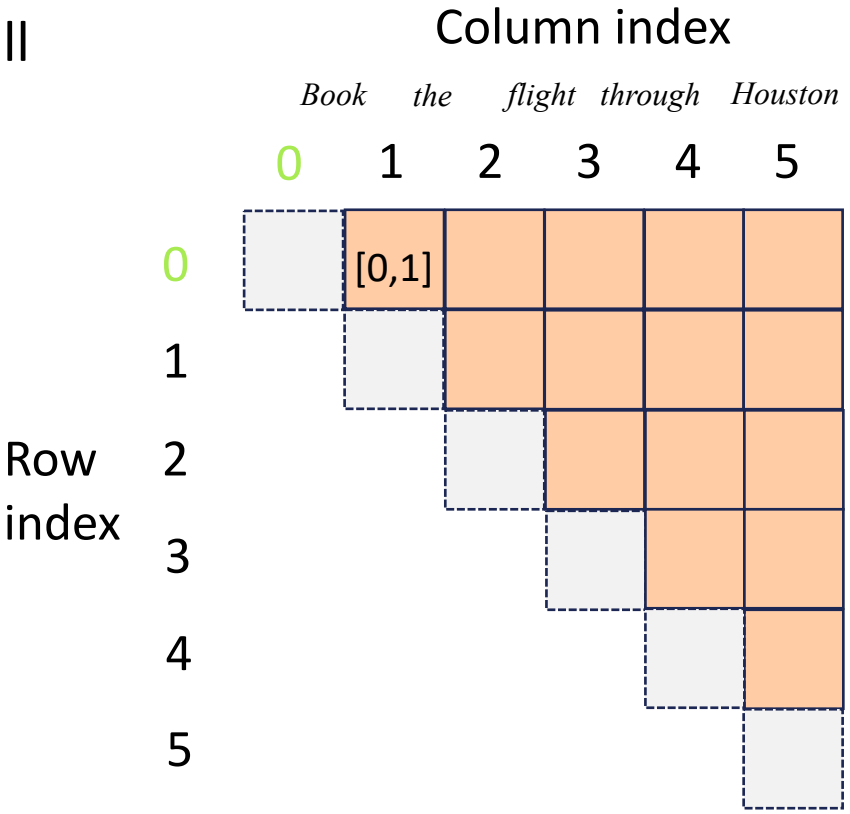
CKY Recognition（识别）

- For a sentence of length n, work with an $(n+1)\times(n+1)$ **matrix** m, using the upper triangular portion ⇒ **parse table**（解析表）
- Cell $m[i,j]$ contains the set of non-terminals representing all constituents that span position i through j.

Index: 0 *Book* 1 *the* 2 *flight* 3 *through* 4 *Houston* 5

Ex. Cell $m[0,3]$ contains the set of non-terminals for all constituents spanning 0 to 3

Goal: Fill in the parse table correctly in a bottom-up way



CKY Recognition: Task breakdown

- Because grammar is in **CNF** ⇒ each non-terminal entry has exactly **two children**.
- For each constituent represented by entry $[i, j]$, there must be a position in the input, k, where it can be split into two parts $(i < k < j)$
- The first component $[i, k]$ is to the left of $[i, j]$, at the same row i
- The second component $[k, j]$ is beneath $[i, j]$, at the same column j
- Then, $[i, j]$ is the **combination** of $[i, k]$ and $[k, j]$ for all possible k

The smallest problems are $[j-1, j]$, i.e., spans of single word

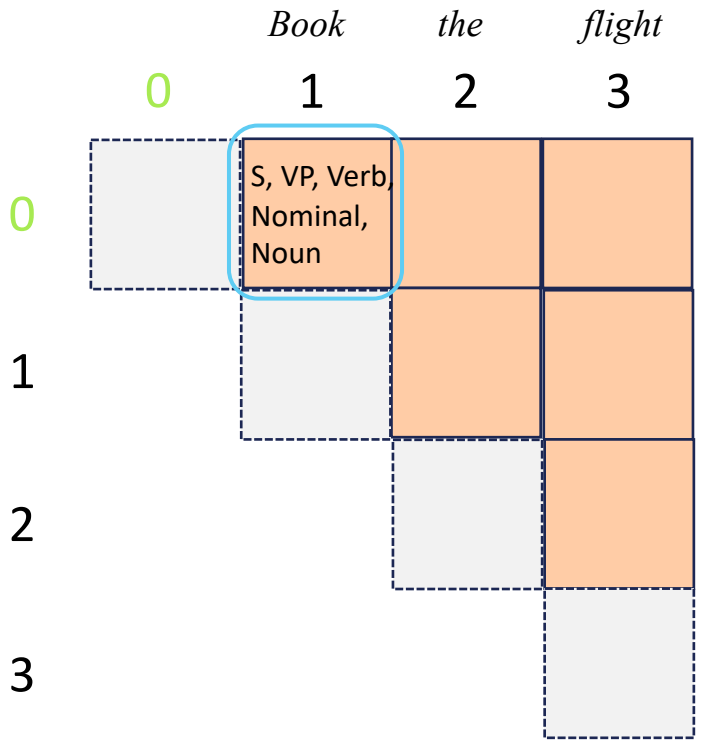
In order to solve the **problem**, we need to solve **smaller problems** first

- A typical dynamic programming (DP) approach

CKY Recognition: Example

Index: 0 *Book* 1 *the* 2 *flight* 3

Solve span [0, 1]



5 production rules match with ***"Book"***

Store them in **cell [0,1]**

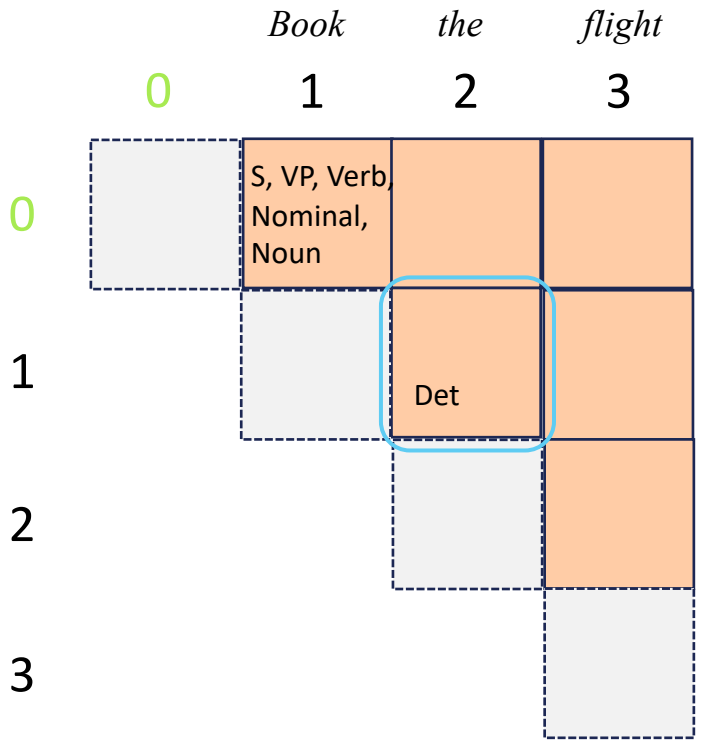
$m[0,1] = \{S, Nominal, VP, Noun, Verb\}$

$\mathcal{L}_1$ in CNF
$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$
$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$

CKY Recognition: Example

Index: 0 *Book* 1 *the* 2 *flight* 3

Solve span [1, 2]



1 production rule matches with ***"the"***

Store it in **cell [1,2]**

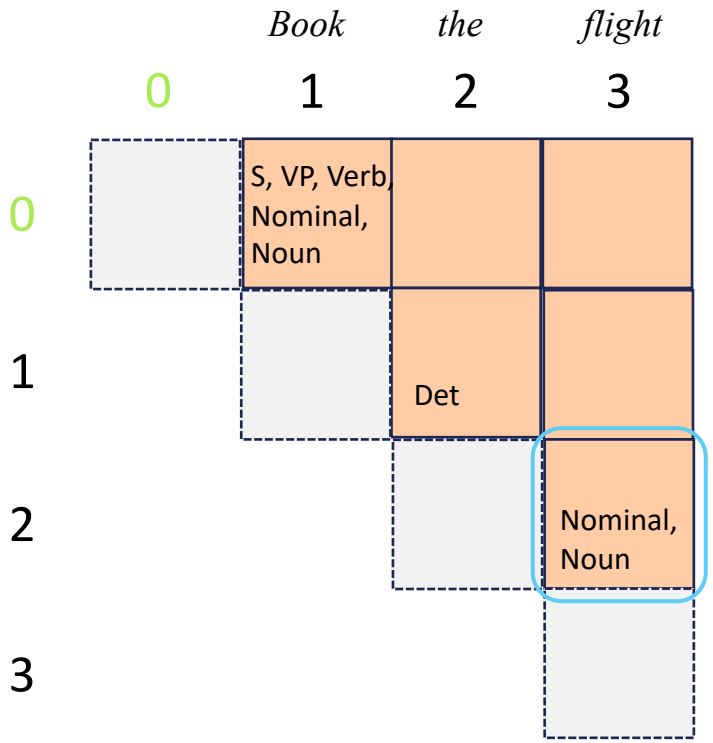
$m[1,2] = \{Det\}$

$\mathcal{L}_1$ in CNF
$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$
$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$

CKY Recognition: Example

Index: 0 *Book* 1 *the* 2 *flight* 3

Solve span [2, 3]



2 production rules match with ***"flight"***

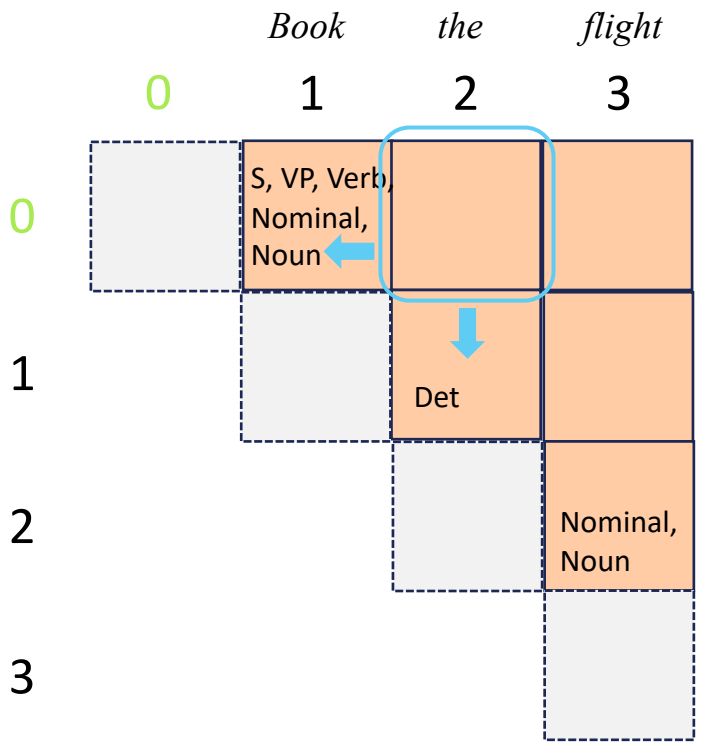
Store them in **cell [2,3]**

$m[2,3]$
$= \{Nominal, Noun\}$

$\mathcal{L}_1$ in CNF
$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$
$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$

CKY Recognition: Example

Index: 0 *Book* 1 *the* 2 *flight* 3



Solve span [0, 2]

Look up the grammar table to find a rule $A \rightarrow B\ C$ such that

$B \in m[0,1]$ and $C \in m[1,2]$

No rule is found, so let

$$m[2,3] = \emptyset$$

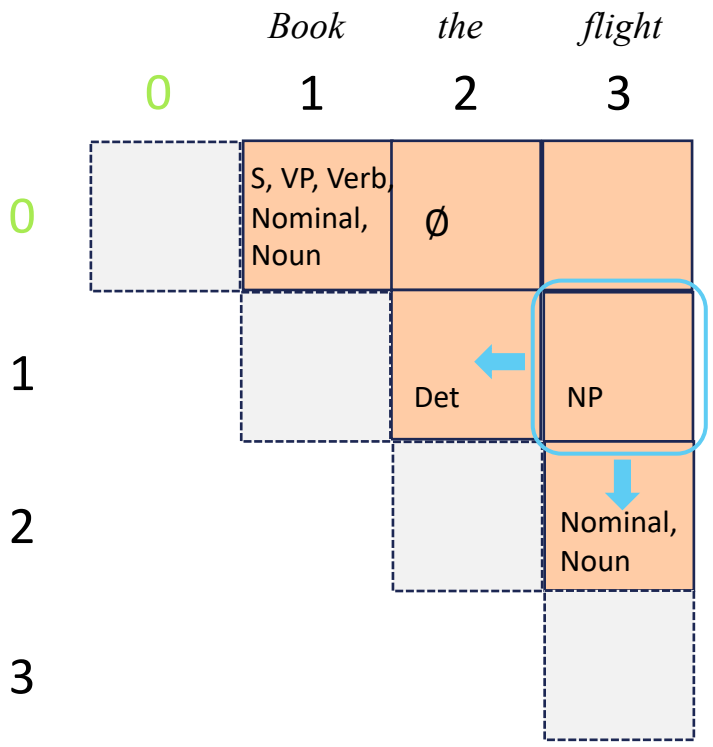
It means the phrase "*Book the*" is incomplete in grammar

（语法不完整）

$\mathcal{L}_1$ in CNF
$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$
$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$

CKY Recognition: Example

Index: 0 *Book* 1 *the* 2 *flight* 3



Solve span [1, 3]

Look up the grammar table to find a rule $A \rightarrow B\ C$ such that
$B \in m[1,2]$ and $C \in m[2,3]$

Found one rule!

$$NP \rightarrow Det\ Nominal$$

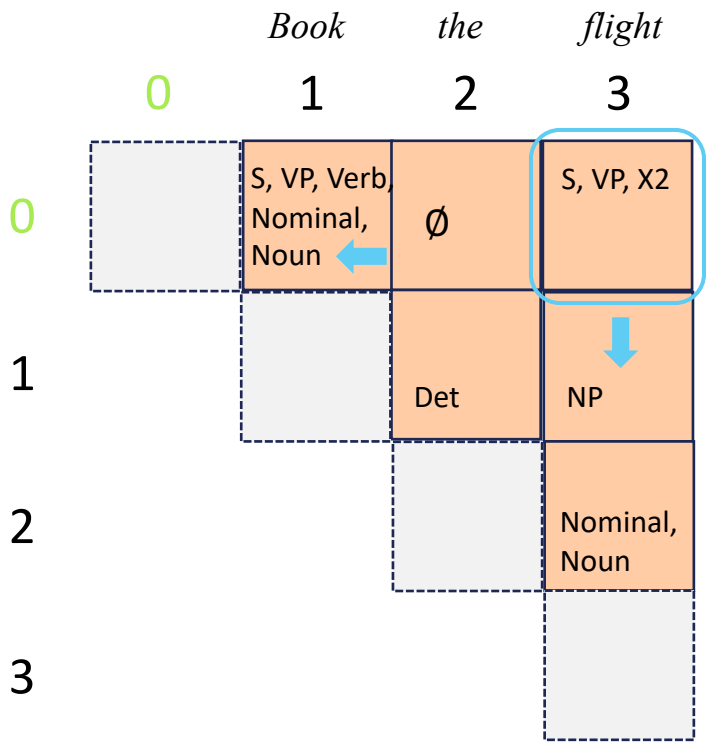
Store it:

$$m[2,3] = \{NP\}$$

$\mathcal{L}_1$ in CNF
$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$
$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$

CKY Recognition: Example

Index: 0 *Book* 1 *the* 2 *flight* 3



Solve span [0, 3]

There are two ways to split:

$[0,3] = [0,1] + [1,3]$,

$[0,3] = [0,2] + [2,3]$

But since $m[0,2] = \emptyset$, only consider the first split

Look up the grammar table to find a rule $A \rightarrow B\ C$ such that

$B \in m[0,1]$ and $C \in m[1,3]$

Found three rules

$$m[0,3] = \{S, VP, X2\}$$

$\mathcal{L}_1$ in CNF
$S \rightarrow NP\ VP$
$S \rightarrow X1\ VP$
$X1 \rightarrow Aux\ NP$
$S \rightarrow book \mid include \mid prefer$
$S \rightarrow Verb\ NP$
$S \rightarrow X2\ PP$
$S \rightarrow Verb\ PP$
$S \rightarrow VP\ PP$
$NP \rightarrow I \mid she \mid me$
$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$
$VP \rightarrow X2\ PP$
$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

$Det \rightarrow that \mid this \mid the \mid a$
$Noun \rightarrow book \mid flight \mid meal \mid money$
$Verb \rightarrow book \mid include \mid prefer$

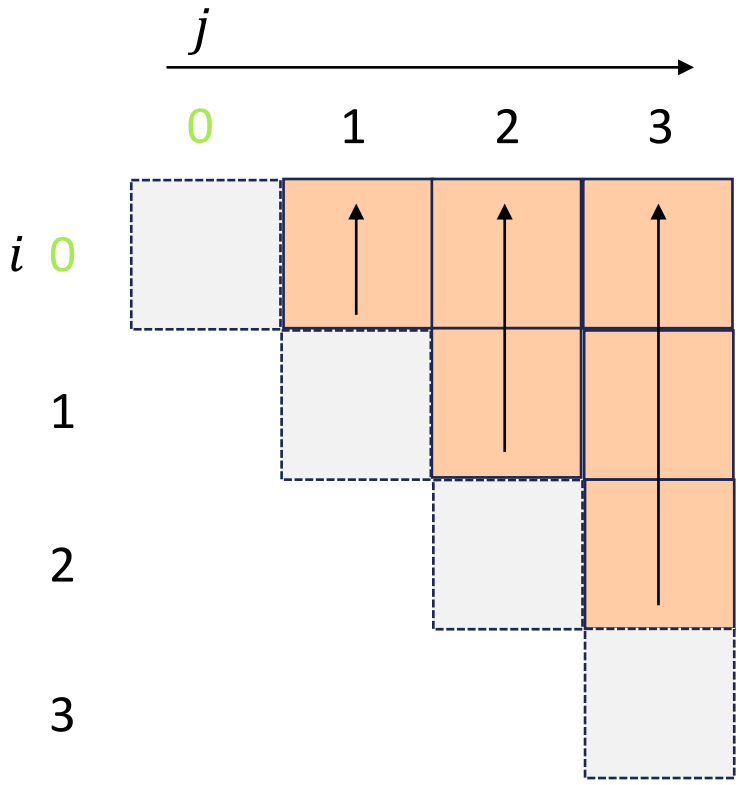
CKY Algorithm

Outer loop: solves the **span** $[0, j]$

```
function CKY-PARSE(words, grammar) returns table

  for j ← from 1 to LENGTH(words) do
      for all {A | A → words[j] ∈ grammar}
              table[j − 1, j] ← table[j − 1, j] ∪ A
      for i ← from j − 2 down to 0 do
          for k ← i + 1 to j − 1 do
              for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
                      table[i,j] ← table[i,j] ∪ A
```

First inner loop: solves the base case **span** $[j-1, j]$



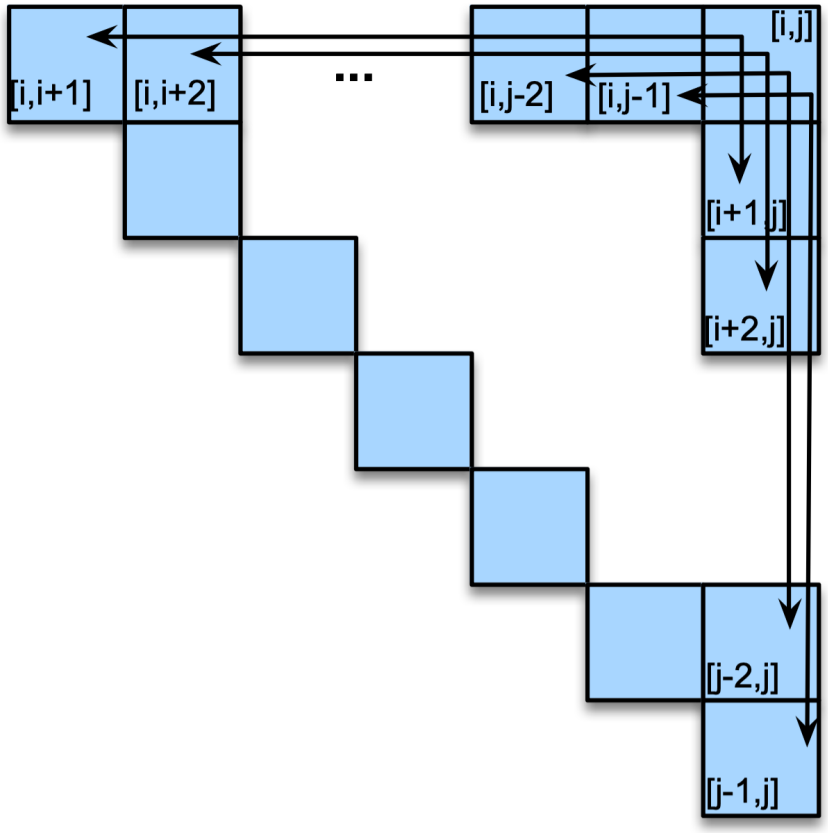
Second inner loop: solves the subproblems ranging from **span** $[j-2, j]$ to **span** $[0, j]$

inner-inner loop: solves each subproblem by breaking it down **span** $[i, j] \Rightarrow$ **span** $[i, k]$ **+** $[k, j]$

CKY Parsing

- The CKY algorithm given in previous slide is a **recognizer**, not a **parser**
- As a parser, it should return all possible parses (trees) for a given input
- Need to make two changes:
- 1. Augment table entries with ***pointers*** (to entries from which the current non-terminal is derived)
- 2. Allow multiple versions of the same non-terminal to be entered
- With these changes, the completed table contains all possible parses for the input
- choose an S from cell $[0, n]$ ⇒ recursively retrieve its constituents from the table

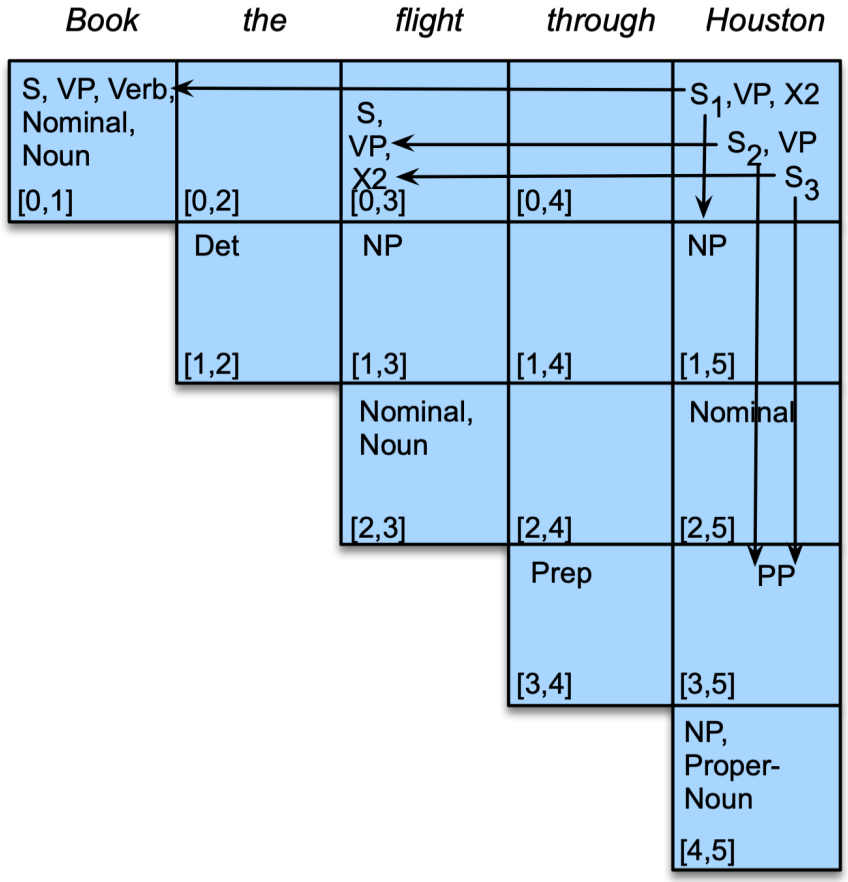
CKY Parsing - Add pointers



Entry $[i, j]$ contains a pair of pointers for each non-terminal in it, which point to the children non-terminals in entry $[i, k]$ and $[k, j]$ $\quad (\mathrm{i} + 1 \leq k \leq j - 1)$

Fig. 17.13., Ch 17, pp 15, SLP3

CKY Parsing - Multiple versions for same non-terminal



Start with cell $[0,5]$, which contains three versions of S non-terminal:

$S_1 \rightarrow Verb\ \ NP$

$S_2 \rightarrow VP\ \ PP$

$S_3 \rightarrow X2\ \ PP$

Follow the pointers recursively to build the parse tree

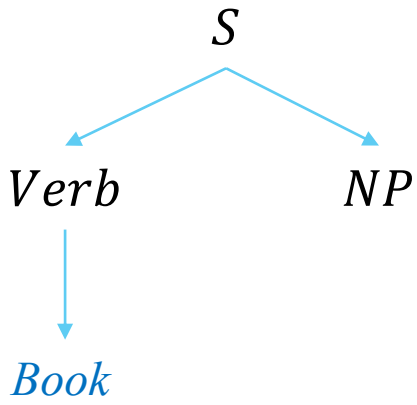


Fig. 17.14., Ch 17, pp 16, SLP3

CKY Parsing - Build the parse tree recursively

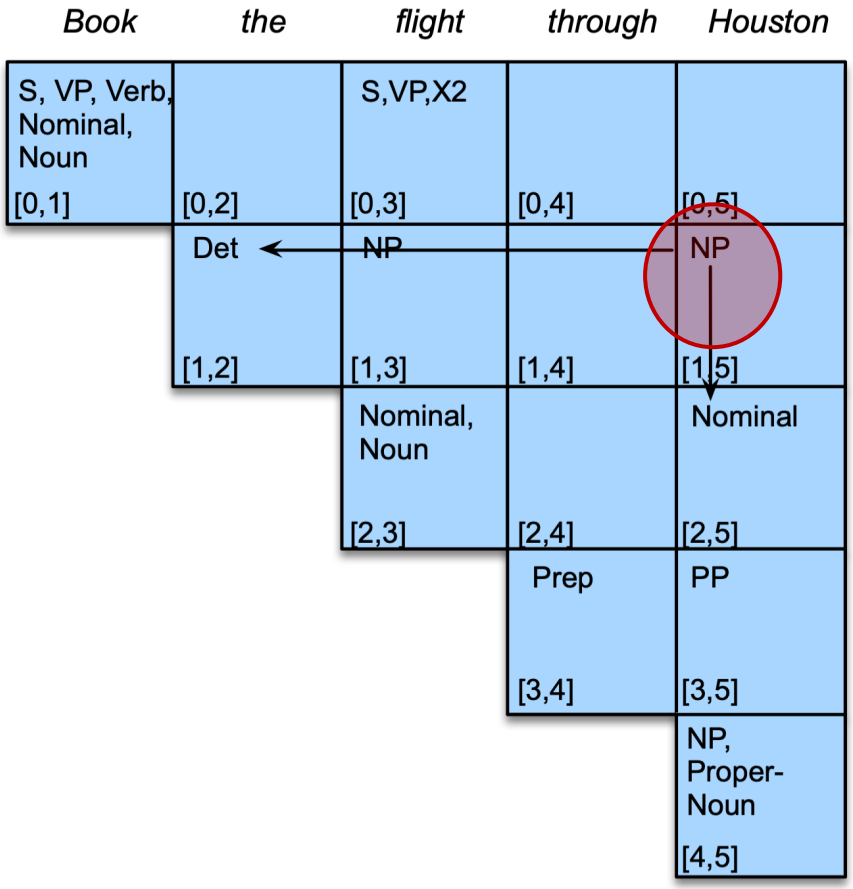
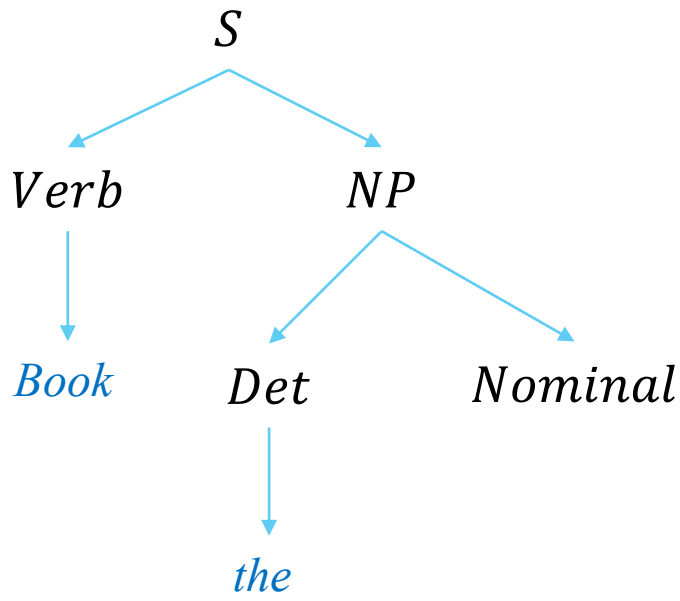


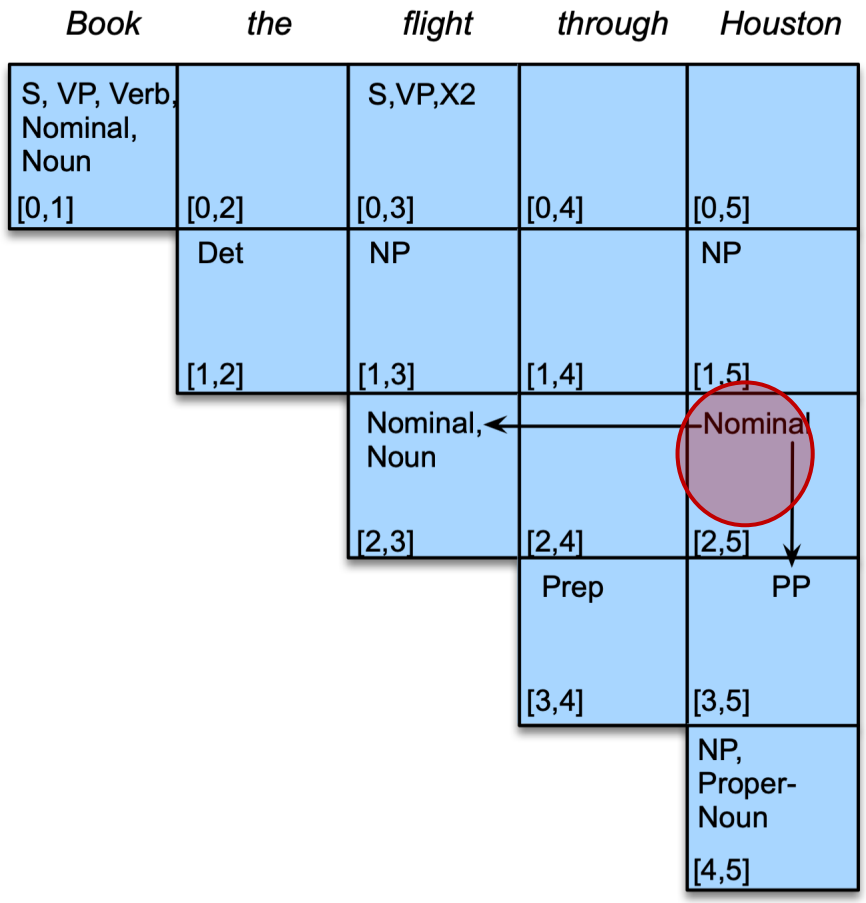
Fig. 17.14., Ch 17, pp 16, SLP3

Go to cell [1,5]

$NP \rightarrow Det \ \ Nominal$



CKY Parsing - Build the parse tree recursively



Go to cell [2,5]

$Nominal \rightarrow Nominal \ \ PP$

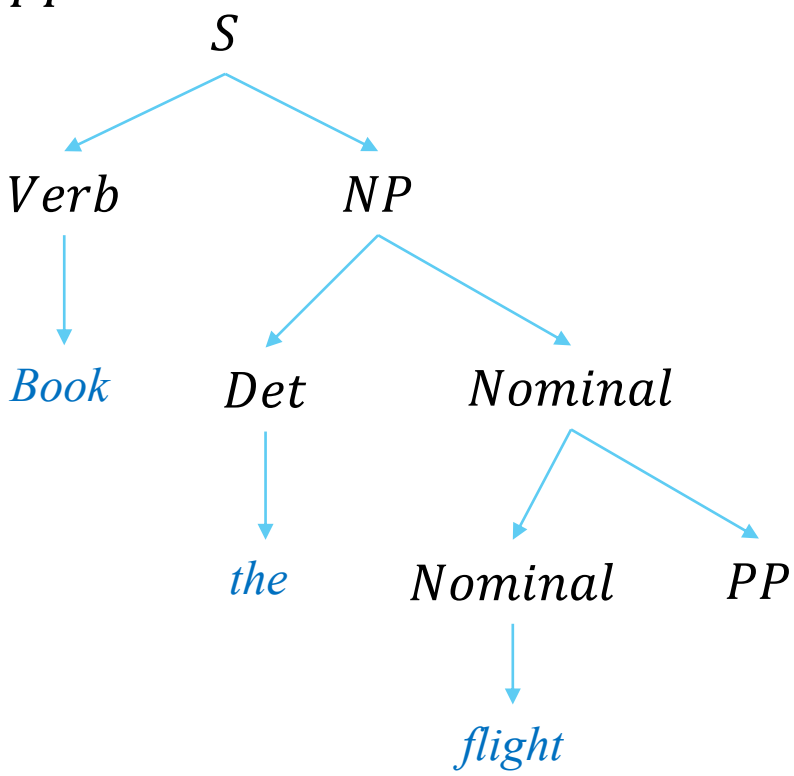
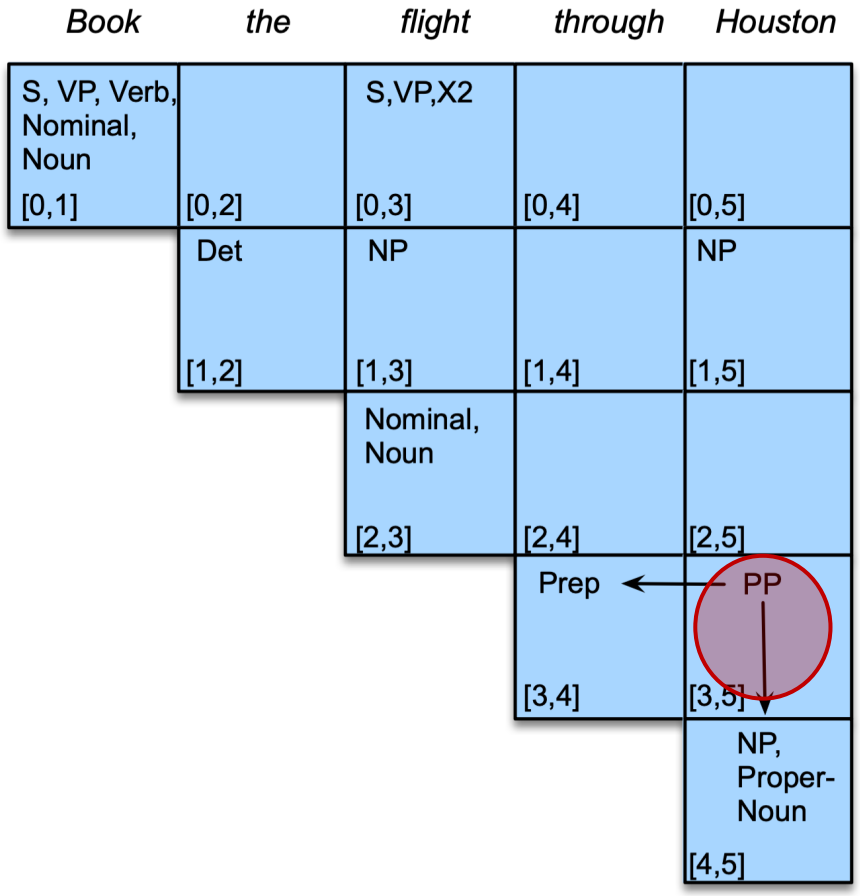


Fig. 17.14., Ch 17, pp 16, SLP3

CKY Parsing - Build the parse tree recursively



Go to cell [3,5]

$PP \rightarrow Prep\ \ NP$

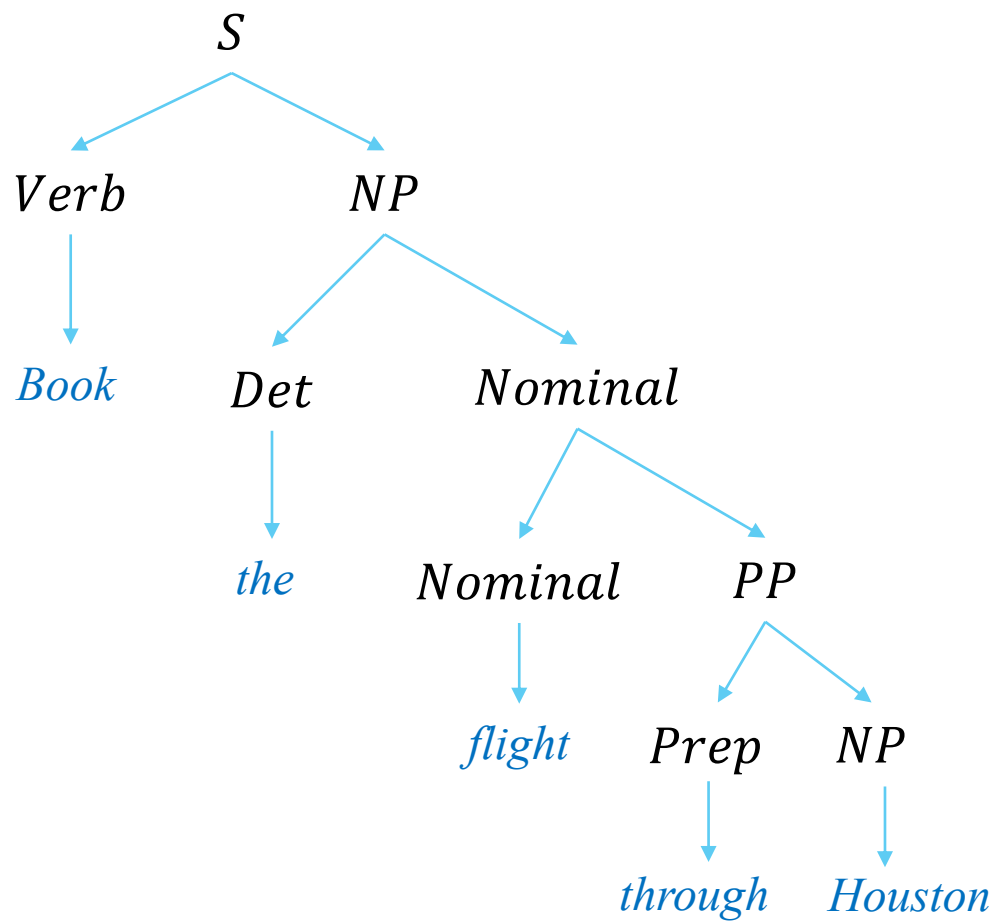


Fig. 17.14., Ch 17, pp 16, SLP3

Basic CKY Parsing's Problem

- It doesn't tell which parse tree is the correct one!

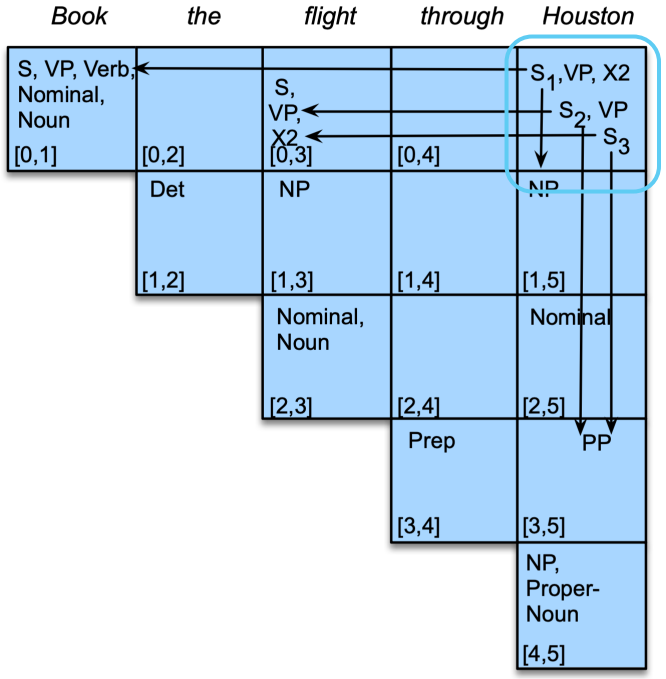


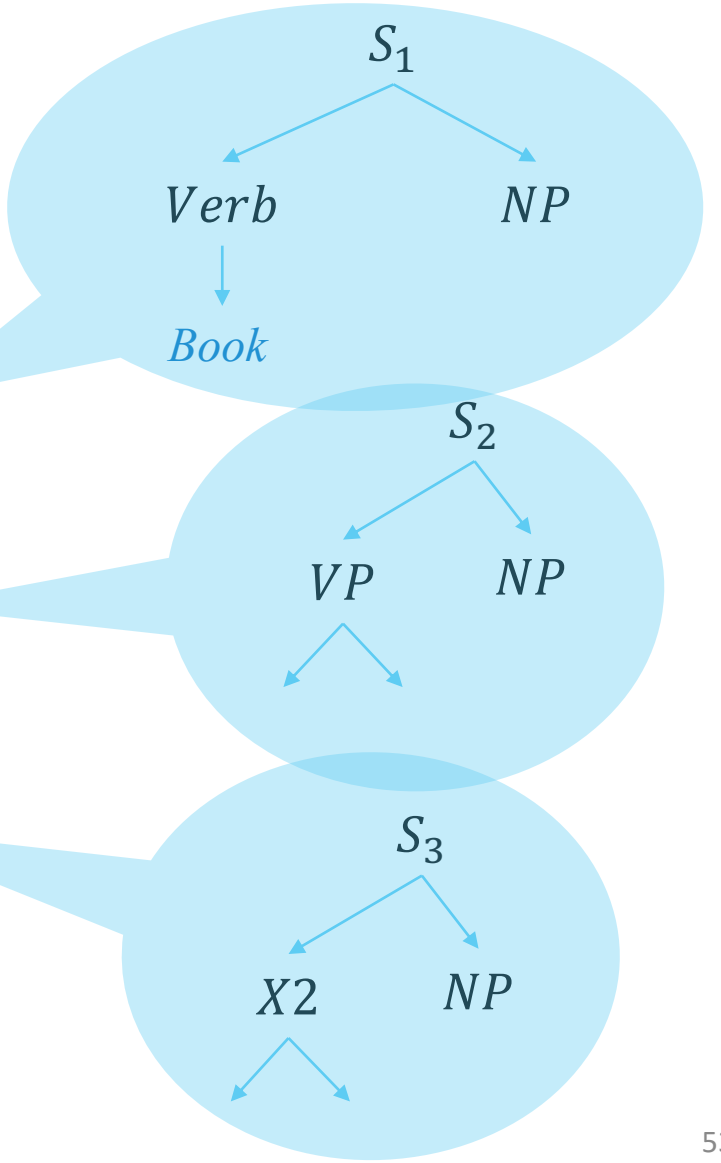
Fig. 17.14., Ch 17, pp 16, SLP3

$S_1 \rightarrow Verb\ \ NP$

$S_2 \rightarrow VP\ \ PP$

$S_3 \rightarrow X2\ \ PP$

These sentences are possible, but not necessarily likely



PCFG Example

A PCFG version to the grammar $\mathcal{L}_1$

Grammar		Lexicon
$S \rightarrow NP\ VP$	[.80]	$Det \rightarrow that$ [.10] | a [.30] | the [.60]
$S \rightarrow Aux\ NP\ VP$	[.15]	$Noun \rightarrow book$ [.10] | $trip$ [.30]
$S \rightarrow VP$	[.05]	| $meal$ [.05] | $money$ [.05]
$NP \rightarrow Pronoun$	[.35]	| $flight$ [.40] | $dinner$ [.10]
$NP \rightarrow Proper\text{-}Noun$	[.30]	$Verb \rightarrow book$ [.30] | $include$ [.30]
$NP \rightarrow Det\ Nominal$	[.20]	| $prefer$ [.40]
$NP \rightarrow Nominal$	[.15]	$Pronoun \rightarrow I$ [.40] | she [.05]
$Nominal \rightarrow Noun$	[.75]	| me [.15] | you [.40]
$Nominal \rightarrow Nominal\ Noun$	[.20]	$Proper\text{-}Noun \rightarrow Houston$ [.60]
$Nominal \rightarrow Nominal\ PP$	[.05]	| NWA [.40]
$VP \rightarrow Verb$	[.35]	$Aux \rightarrow does$ [.60] | can [.40]
$VP \rightarrow Verb\ NP$	[.20]	$Preposition \rightarrow from$ [.30] | to [.30]
$VP \rightarrow Verb\ NP\ PP$	[.10]	| on [.20] | $near$ [.15]
$VP \rightarrow Verb\ PP$	[.15]	| $through$ [.05]
$VP \rightarrow Verb\ NP\ NP$	[.05]	
$VP \rightarrow VP\ PP$	[.15]	
$PP \rightarrow Preposition\ NP$	[1.0]	

Probabilities are estimated based on a corpus

- These numbers are for pedagogical purposes
- Any real corpus would have more rules and the real probabilities will be much smaller

Use PCFG for disambiguating parsing results

- The probability of a particular parse tree T is defined as the product of probabilities of all the n rules to expand each of the n non-terminals

$$p(T) = \prod_{i=1}^{n} p(A_i \rightarrow \beta_i)$$

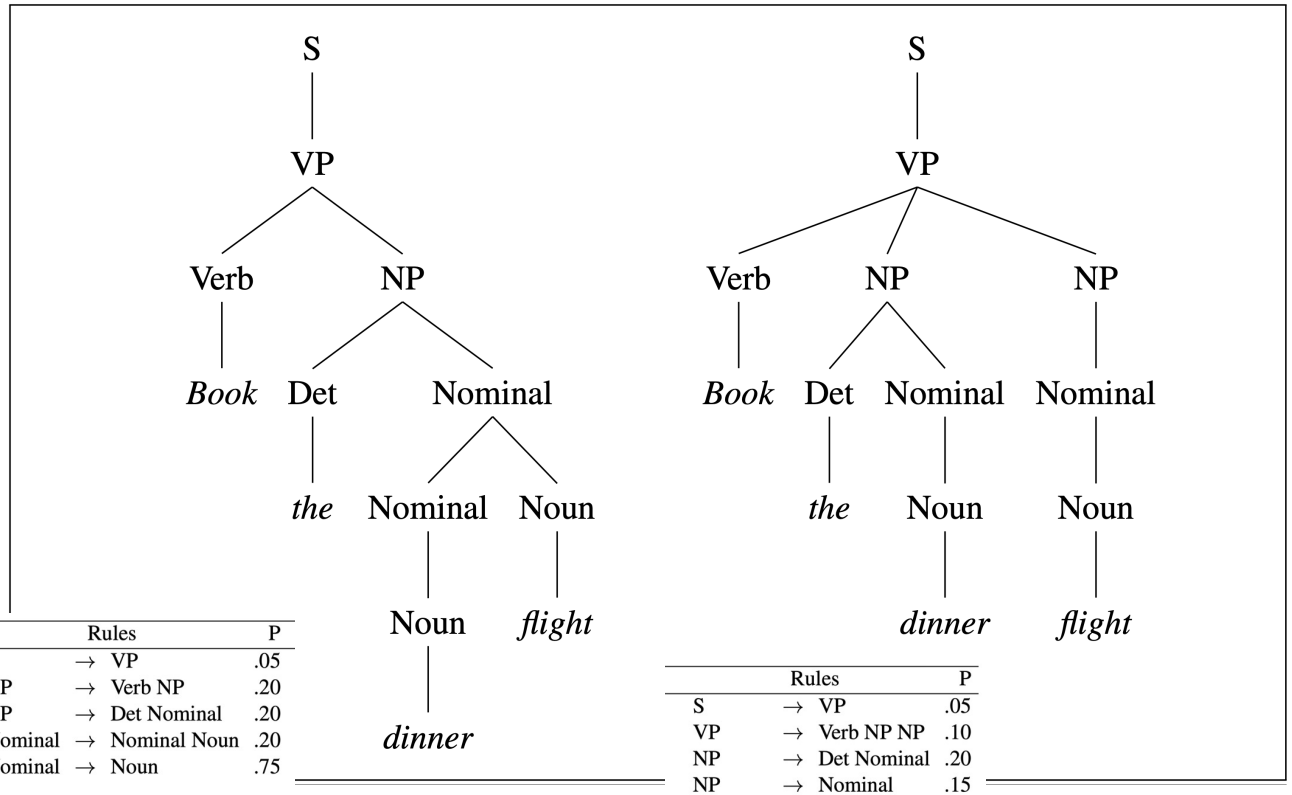
- Consider all possible parse trees of a sentence S, we need to pick the tree with the highest probability $\hat{T}(S) = \arg\max_T p(T)$

PCFG Usage Example

Sentence: *Book the dinner flight*

two senses:

- "Book a flight that serves dinner"
- "Book a flight on behalf of 'the dinner'"



	Rules		P
S	→	VP	.05
VP	→	Verb NP	.20
NP	→	Det Nominal	.20
Nominal	→	Nominal Noun	.20
Nominal	→	Noun	.75
Verb	→	book	.30
Det	→	the	.60
Noun	→	dinner	.10
Noun	→	flight	.40

	Rules		P
S	→	VP	.05
VP	→	Verb NP NP	.10
NP	→	Det Nominal	.20
NP	→	Nominal	.15
Nominal	→	Noun	.75
Nominal	→	Noun	.75
Verb	→	book	.30
Det	→	the	.60
Noun	→	dinner	.10
Noun	→	flight	.40

Compute using the probabilities of PCFG $\mathcal{L}_0$:

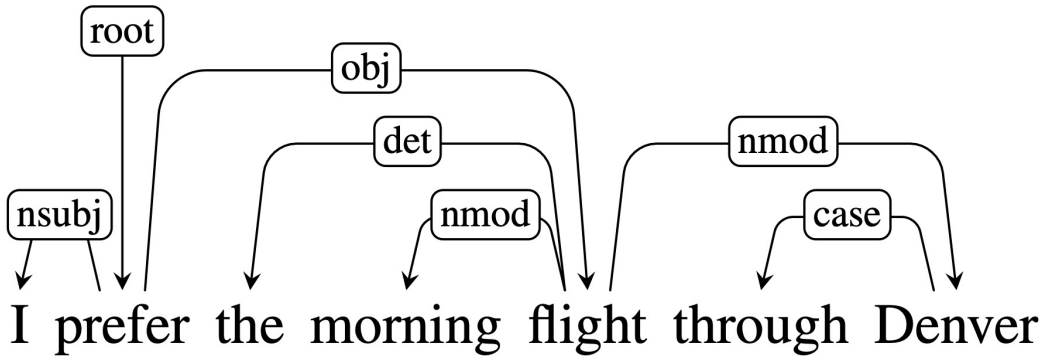
$$P(T_{\text{left}}) = .05 * .20 * .20 * .20 * .75 * .30 * .60 * .10 * .40 = \mathbf{2.2\times10^{-6}}$$



$$P(T_{\text{right}}) = .05 * .10 * .20 * .15 * .75 * .75 * .30 * .60 * .10 * .40 = \mathbf{6.1\times10^{-7}}$$

Dependency Grammars

- Different from context-free grammars and constituency-based representations
- **Directed grammatical relations** between words define sentence structure.



Labeled arcs from **heads** to **dependents**

arc: a directed link

$prefer \xrightarrow{nsubj} I$ *prefer* is the head of *I*

$prefer \xrightarrow{obj} flight$ *flight* is the dependent of *prefer*

$root \rightarrow prefer$ *root* is the head of *prefer* and also the head of the entire structure (root of the tree)

Overview

- Neural Basis of Language Processing
- Part-of-Speech Tagging
- Syntactic Parsing
- **Cognitive Processing**

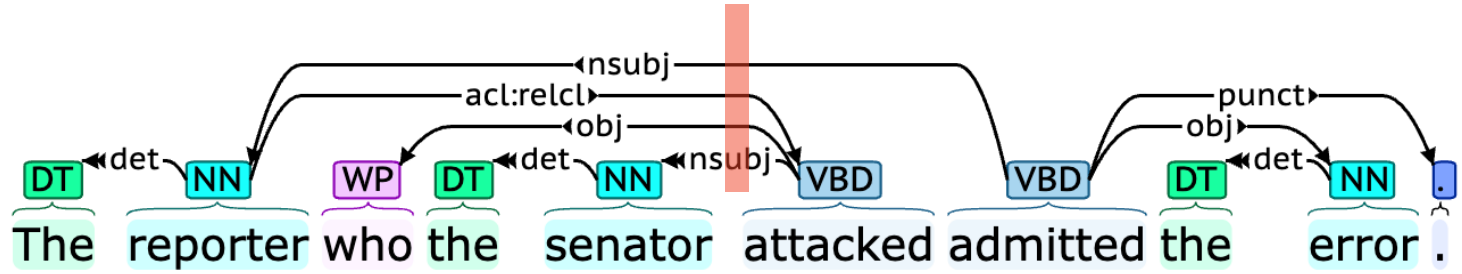
Source: Gibson (2000), https://doi.org/10.1016/S0010-0277(98)00034-1

Cognitive Effort in Processing Language

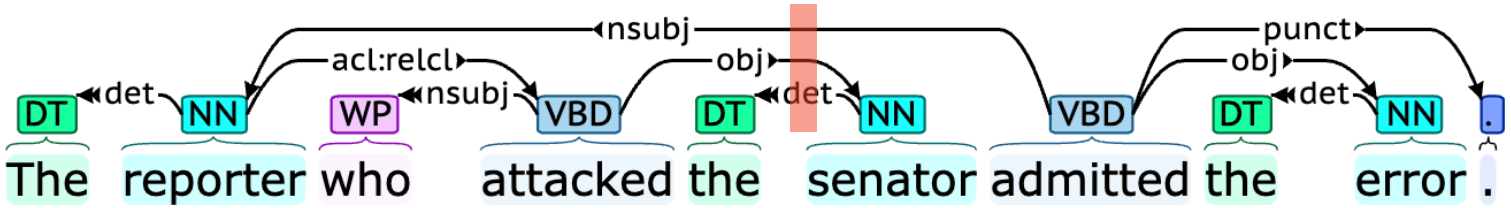
- Theory: **Locality** of syntactic dependencies
- `0: The reporter admitted the error.`
- `1a: The reporter who the senator attacked admitted the error.`
- `1b: The reporter who attacked the senator admitted the error.`
- Both relative clauses (RCs)
  - 1a => “who” is extracted from the **object** position of RC. => Hard
  - 1b => “who” is extracted from the **subject** position of RC. => Easy

Dependency parsed sentences

- 1a: The reporter who the senator attacked admitted the error.



more stacked arrows



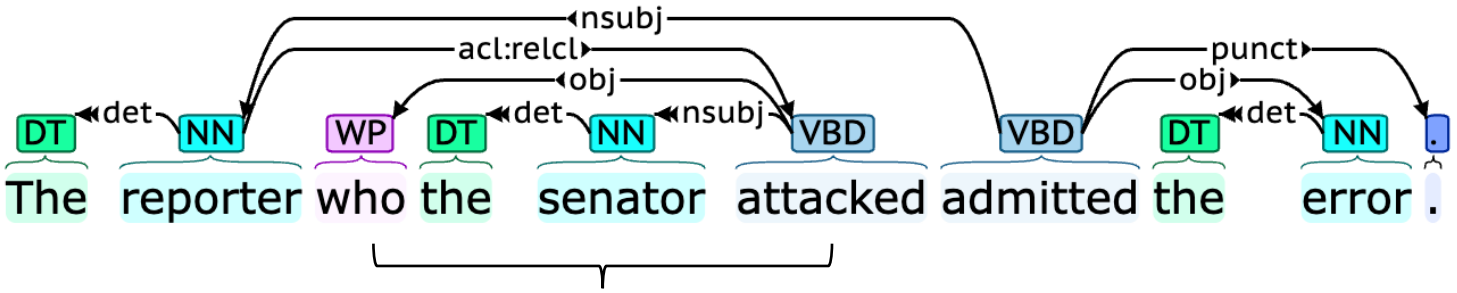
fewer stacked arrows

- 1b: The reporter who attacked the senator admitted the error.

The memory costs are different

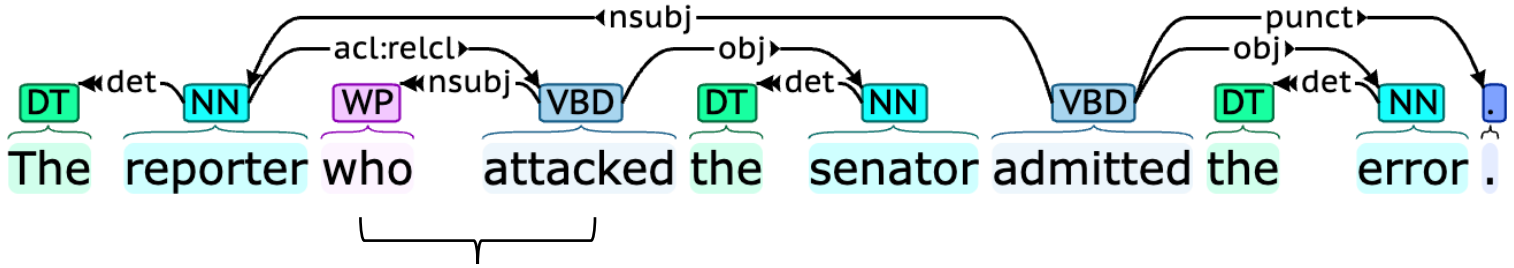
- One explanation: the distance between “who” and “attacked” determines the memory cost to resolve their dependency relation -- locality.

1a



less “local”, more memory cost to wait for “the senator” to be parsed.

1b



more “local”, less memory cost

Cognitive Effort in Processing Language

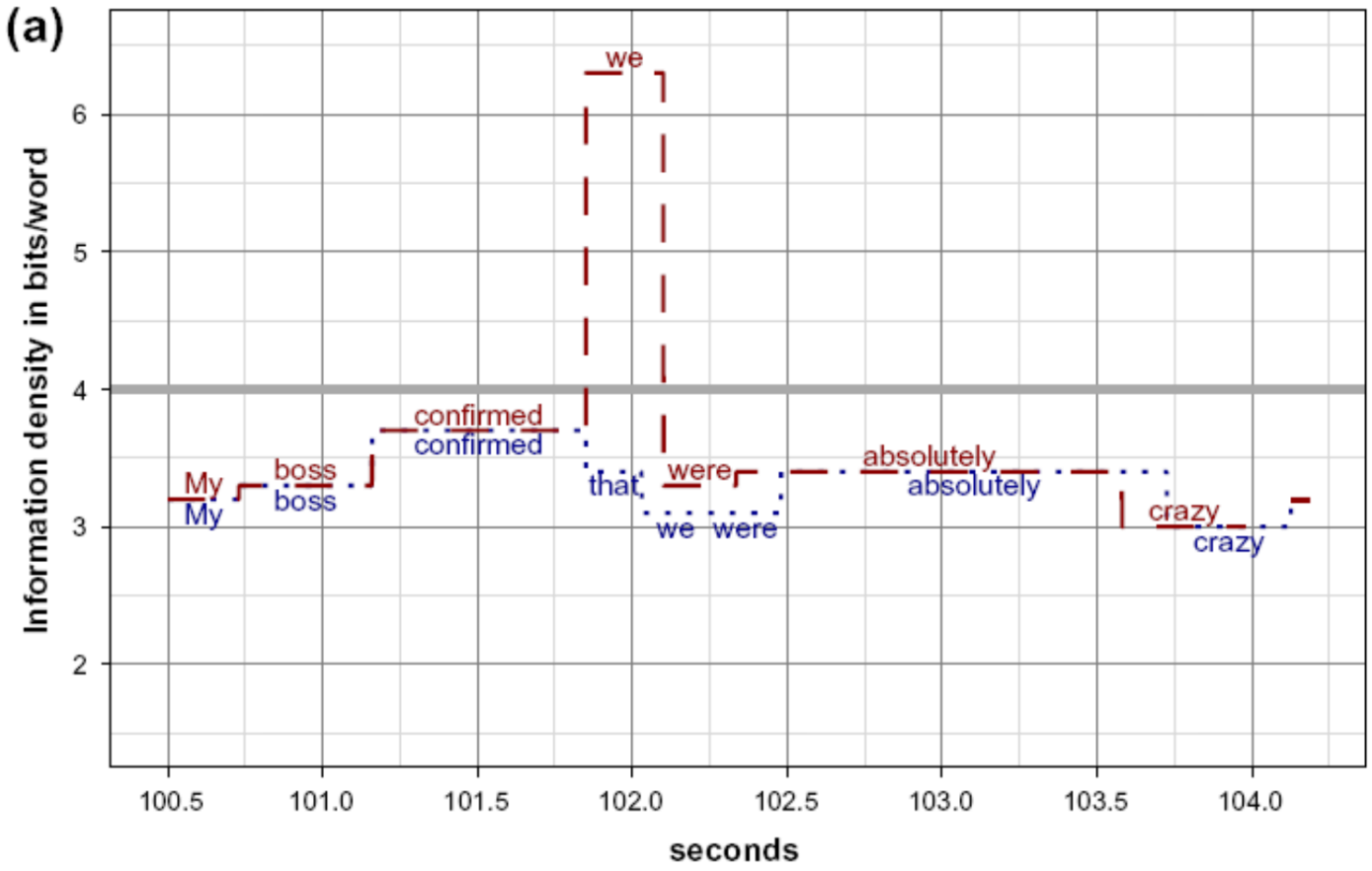
- Theory: Uniform information density (**UID**).
  - Human speakers tend to distribute the amount of information uniformly in natural language.
- `1a: My boss confirmed we were absolutely crazy.`
- `1b: My boss confirmed that we were absolutely crazy.`

- Which one reads more smoothly?

Source: Jaeger 2010, doi:10.1016/j.cogpsych.2010.02.002

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Uniform Information Density



Human speakers tend to avoid the sudden changes in information.